

Phase-1 Submission Template

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1.Problem Statement;

The rapid spread of fake news on digital platforms poses a significant threat to societal trust, public safety, and democratic processes. This project aims to build an NLP-based system that can automatically detect fake news, helping users and organizations make informed decisions and combat misinformation effectively.

2.Objectives of the Project;

- *interface. Build a machine learning model that accurately classifies news as real or fake.*
- *Analyze linguistic patterns in fake vs real news using NLP techniques.*
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3.Scope of the Project;

- *Text preprocessing and NLP-based feature extraction*
- *Multiple classification models (Logistic Regression, Random Forest, etc.)*

Limitations/Constraints:

- *Model trained on static datasets*
- *Focus limited to English-language news.*

4.Data Sources;

- *Fake and real news dataset from Kaggle (e.g., “Fake News Detection” dataset)*
- *Static dataset downloaded once.*

Data Source Link: <https://www.kaggle.com/datasets/mahdimashayekhi/fake-news-detection-dataset>

5.High-Level Methodology;

- *Data Collection: Download dataset from Kaggle.*
- *Data Cleaning: Remove duplicates, handle missing values, standardize text formatting.*
- *EDA: Word clouds, frequency plots, sentiment analysis.*
- *Feature Engineering: TF-IDF, n-grams, stop word removal, POS tagging.*
- *Model Building: Logistic Regression, Naive Bayes, Random Forest, XG Boost.*
- *Model Evaluation: Accuracy, Precision, Recall, F1-score, ROC-AU*
- *Data Collection: Download dataset from Kaggle.*
- *Data Cleaning: Remove duplicates, handle missing values, standardize text formatting.*
- *EDA: Word clouds, frequency plots, sentiment analysis.*
- *Feature Engineering: TF-IDF, n-grams, stop word removal, POS tagging.*
- *Model Building: Logistic Regression, Naive Bayes, Random Forest, XG Boost.*

6.Tools and Technology;

- *Programming Language: Python.*
- *Notebook/IDE: Google Colab, Jupyter Notebook*
- *Libraries: pandas, numpy, matplotlib, seaborn, scikit-learn, nltk, spaCy, TensorFlow/Keras*
- *Deployment Tools (Optional): Streamlit, Gradio, Flask.,*

7.Team Members and Roles;

NAME	ROLE	WORK
MURALIDHARAN.K		<i>Model development and evaluation.</i>
GOWTHAM.P		<i>Data collection, cleaning, and preprocessing.</i>
PUGAZHENTHI.G		<i>Visualization and interpretation of results.</i>
BHARATHIDHASAN.M		<i>(Optional) Deployment and</i>

		<i>UI design.</i>
JESLIN SAJAN.		